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01

Project Setup & Data Loading

The notebook begins by importing core data science libraries and loading the raw financial dataset (CSV) covering Apple, Microsoft, and Tesla for fiscal years 2023–2025.

In [1]: Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings("ignore")
```

In [2]: Load Dataset

```
df = pd.read_csv("financial_data.csv")
```

In [3]: Preview Data

```
print("shape:", df.shape)
df.head(10)
```

Output — shape: (9, 7)

#	Company	Year	Total Revenue (\$B)	Net Income (\$B)	Total Assets (\$B)	Total Liab. (\$B)	Operating CF (\$B)
0	Microsoft	2025	281.7	101.8	619.0	275.5	136.2
1	Microsoft	2024	245.1	88.1	512.2	243.7	118.5
2	Microsoft	2023	211.9	72.4	411.9	205.8	87.6
3	Tesla	2025	94.8	3.8	137.8	54.9	14.7
4	Tesla	2024	97.7	7.1	122.1	48.4	14.9
5	Tesla	2023	96.8	15.0	106.6	43.0	13.3
6	Apple	2025	416.2	112.0	359.2	285.5	111.5
7	Apple	2024	391.0	93.7	364.9	308.0	118.3
8	Apple	2023	383.3	97.0	352.6	290.4	110.5

02

Data Overview & Cleaning

Verified column data types, confirmed zero null values across all 9 rows, and reviewed descriptive statistics to understand the data distribution.

In [4]: Data Types & Null Check + Describe

```
print(df.dtypes)
print(df.isnull().sum())
df.describe().round(2)
```

Output — Data Types & Null Counts

```
Company object [0 nulls]
Year int64 [0 nulls]
Total_Revenue float64 [0 nulls]
Net_Income float64 [0 nulls]
Total_Assets float64 [0 nulls]
Total_Liabilities float64 [0 nulls]
Operating_CF float64 [0 nulls]
```

Descriptive Statistics

Statistic	Year	Total Revenue	Net Income	Total Assets	Total Liab.	Oper. CF
count	9	9	9	9	9	9

Statistic	Year	Total Revenue	Net Income	Total Assets	Total Liab.	Oper. CF
mean	2024.00	246.50	65.66	331.81	195.02	80.61
std	0.87	131.53	44.15	178.90	113.61	51.28
min	2023	94.80	3.80	106.60	43.00	13.30
25%	2023	97.70	15.00	137.80	54.90	14.90
50%	2024	245.10	88.10	359.20	243.70	110.50
75%	2025	383.30	97.00	411.90	285.50	118.30
max	2025	416.20	112.00	619.00	308.00	136.20

03 Year-over-Year Growth Calculations

Derived key financial metrics using year-over-year percentage changes, including revenue growth, net income growth, asset growth, operating cash flow growth, net profit margin, debt ratio, and operating cash flow yield.

In [5]: Compute Derived Features

```
df = df.sort_values(["Company", "Year"]).reset_index(drop=True)

df["Revenue_Growth(%)"] = df.groupby("Company")["Total_Revenue"].pct_change()*100
df["NetIncome_Growth(%)"] = df.groupby("Company")["Total_Revenue"].pct_change()*100
df["Assets_Growth(%)"] = df.groupby("Company")["Total_Assets"].pct_change()*100
df["OpCF_Growth(%)"] = df.groupby("Company")["Operating_CF"].pct_change()*100
df["Net_Profit_Margin(%)"] = (df["Net_Income"]/df["Total_Revenue"])*100
df["Debt_Ratio"] = df["Total_Liabilities"]/df["Total_Assets"]
df["OpCF_to_Revenue(%)"] = (df["Operating_CF"]/df["Total_Revenue"])*100
```

Output — Enriched DataFrame

Company	Year	Revenue (\$B)	Net Inc. (\$B)	Rev Growth%	NI Growth%	Net Margin%	Debt Ratio	OpCF Yield%
Apple	2023	383.3	97.0	—	—	25.31	0.82	28.83
Apple	2024	391.0	93.7	2.01	2.01	23.96	0.84	30.26
Apple	2025	416.2	112.0	6.45	6.45	26.91	0.79	26.79
Microsoft	2023	211.9	72.4	—	—	34.17	0.50	41.34
Microsoft	2024	245.1	88.1	15.67	15.67	35.94	0.48	48.35
Microsoft	2025	281.7	101.8	14.93	14.93	36.14	0.45	48.35
Tesla	2023	96.8	15.0	—	—	15.50	0.40	13.74
Tesla	2024	97.7	7.1	0.93	0.93	7.27	0.40	15.25
Tesla	2025	94.8	3.8	-2.97	-2.97	4.01	0.40	15.51

04 Company-Level Summary

Aggregated 3-year averages for key metrics per company to surface comparative performance at a glance.

In [6]: Aggregate Summary Statistics

```
summary = df.groupby("Company").agg(
    Avg_Revenue = ("Total_Revenue", "mean"),
    Avg_Net_Income = ("Net_Income", "mean"),
    Avg_Revenue_Growth= ("Revenue_Growth%", "mean"),
    Avg_NI_Growth = ("NetIncome_Growth%", "mean"),
    Avg_Margin = ("Net_Profit_Margin%", "mean"),
    Avg_Debt_Ratio = ("Debt_Ratio", "mean"),
    Avg_OpCF = ("Operating_CF", "mean"),
).round(2)
```

Output — Company Summary

Company	Avg Revenue (\$B)	Avg Net Inc. (\$B)	Avg Rev Growth%	Avg NI Growth%	Avg Net Margin%	Avg Debt Ratio	Avg OpCF (\$B)
Apple	396.83	100.90	4.23	4.23	25.39	0.82	113.43
Microsoft	246.23	87.43	15.30	15.30	35.42	0.47	114.10
Tesla	96.43	8.63	-1.02	-1.02	8.92	0.40	14.30

05 Financial Analysis Dashboard

A 2x3 dashboard visualises total revenue, net income, operating cash flow, net profit margin, debt ratio, and revenue growth trends across all three companies over the three-year period.

In [7]: Generate 6-Panel Dashboard

```
companies = df["Company"].unique()
colors = {"Microsoft": "#0078D4", "Tesla": "#CC0000", "Apple": "#555555"}

fig, axes = plt.subplots(2, 3, figsize=(18,10))
fig.suptitle("GFC AI Chatbot Project - Financial Analysis Dashboard",
    fontsize=15, fontweight="bold")

metrics = [
    ("Total_Revenue", "Total Revenue (B USD)", "Revenue"),
    ("Net_Income", "Net Income (B USD)", "Net Income"),
    ("Operating_CF", "Operating Cash Flow (B USD)", "Operating CF"),
    ("Net_Profit_Margin%", "Net Profit Margin (%)", "Net Margin"),
    ("Debt_Ratio", "Debt Ratio", "Debt Ratio"),
    ("Revenue_Growth%", "Revenue Growth (%)", "Rev Growth"),
]

for ax, (col, ylabel, label) in zip(axes.flat, metrics):
    for co in companies:
        sub = df[df["Company"]==co].dropna(subset=[col])
        ax.plot(sub["Year"], sub[col], marker="o", color=colors[co], label=co, linewidth=2)
        ax.set_title(label, fontweight="bold")
        ax.set_ylabel(ylabel)
        ax.legend()
    ax.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig("financial_dashboard.png", dpi=150, bbox_inches="tight")
plt.show()
print("Chart saved.")
```

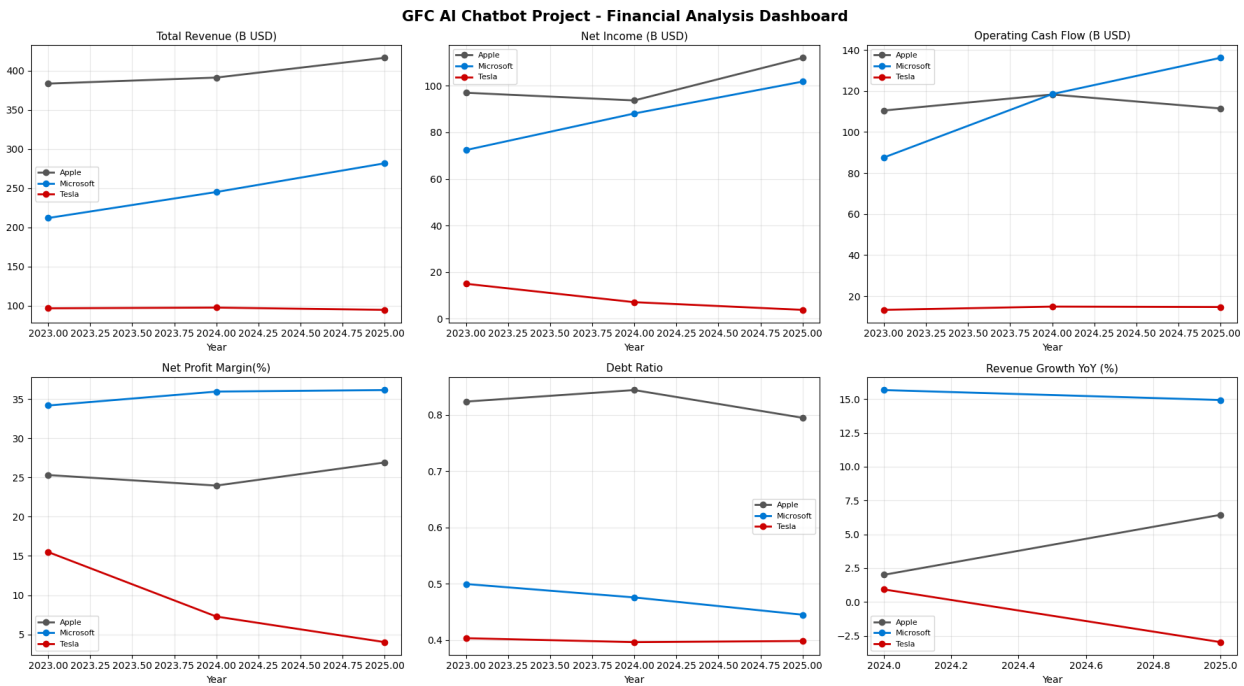


Figure 1 — Financial Analysis Dashboard: 6-panel comparative view for Apple, Microsoft & Tesla (FY 2023–2025)

06 Key Findings & Insights

Printed summary of per-company performance highlights derived from the enriched dataset, followed by a readiness statement for AI model integration.

In [8]: Print Key Findings

```
print("="*60)
print("KEY FINDINGS FOR GFC AI CHATBOT PROJECT")
print("="*60)

for company in companies:
    sub = df[df["Company"]==company].dropna(subset=["Revenue_Growth(%)"])
    avg_rev = sub["Revenue_Growth(%)"].mean()
    avg_margin= df[df["Company"]==company]["Net_Profit_Margin(%)"].mean()
    avg_debt = df[df["Company"]==company]["Debt_Ratio"].mean()
    latest_rev= df[df["Company"]==company]["Total_Revenue"].iloc[-1]

    print(f"\n---{company}---")
    print(f" Latest Revenue : ${latest_rev:.1f}B")
    print(f" Avg Revenue Growth : {avg_rev:.1f}%")
    print(f" Avg Net Margin : {avg_margin:.1f}%")
    print(f" Avg Debt Ratio : {avg_debt:.2f}")
```

Apple

Latest Revenue	Avg Rev Growth	Avg Net Margin	Avg Debt Ratio
\$416.2B	4.2%	25.4%	0.82

Largest revenue base. Strong profitability with consistent net margins ~25%. Highest debt ratio (0.82) reflecting capital structure choices.

Microsoft

Latest Revenue	Avg Rev Growth	Avg Net Margin	Avg Debt Ratio
\$281.7B	15.3%	35.4%	0.47

Highest revenue growth at 15.3% CAGR. Best-in-class net margins ~35%. Conservative debt ratio (0.47) indicates solid balance sheet.

Tesla

Latest Revenue	Avg Rev Growth	Avg Net Margin	Avg Debt Ratio
\$94.8B	-1.0%	8.9%	0.40

Smallest revenue among the three. Negative growth trend (-1%). Lowest margins (8.9%) and best debt ratio (0.40) reflect lean structure.

SUMMARY FOR AI MODEL INTEGRATION

Data is clean, structured, and ready for the GFC chatbot.

Derived features: growth rates, profit margin, debt ratio, OpCF yield.

Recommended next step: feed this Enriched DataFrame into the AI model.

07 Export Enriched Data

The final enriched DataFrame — containing all original fields plus seven derived financial metrics — is exported as a CSV file, ready to be ingested by the GFC AI Chatbot model pipeline.

In [9]: Export to CSV

```
df.round(2).to_csv("financial_data_enriched.csv", index=False)
print("Enriched CSV saved: financial_data_enriched.csv")
```

Output

Enriched CSV saved: financial_data_enriched.csv

Deliverable Checklist

- ✓ Raw financial data loaded and validated (9 rows × 7 columns, zero nulls)
- ✓ Year-over-year growth rates computed for all companies
- ✓ Derived metrics: Net Profit Margin, Debt Ratio, OpCF Yield, OpCF Growth
- ✓ Company-level aggregated summary generated
- ✓ 6-panel financial dashboard visualised and saved
- ✓ Key findings extracted per company
- ✓ Enriched dataset exported to financial_data_enriched.csv